

Γ2018, Fall.

1). Since $A^2x = A(Ax) = 0$ for all $x \in \mathbb{R}^n$, $\text{Im } A \subseteq \ker A$. Therefore,

$$\text{rank } A = n - \dim \ker A \leq n - \text{rank } A \Rightarrow 2 \cdot \text{rank } A \leq n$$

2). Since $\det(A - tI) = \det \begin{pmatrix} -t & 0 & -2 \\ 1 & 2-t & 1 \\ 1 & 0 & 3-t \end{pmatrix} = \det \begin{pmatrix} 0 & -t & -2 \\ 2-t & 1 & 1 \\ 0 & 1 & 3-t \end{pmatrix}$

$$= (2-t)(t(t-3)+2) = (2-t)(t^2-3t+2) = (2-t)(t-2)(t-1) = (t-2)^2(t-1)$$

$$\ker(A - 2I) = \ker \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \text{Span}_{\mathbb{R}} \{ (0, 1, 0), (1, 0, -1) \}.$$

$$\ker(A - I) = \ker \begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} = \text{Span}_{\mathbb{R}} \{ (2, -1, -1) \}.$$

Thus it is diagonalizable, with $U = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 0 & -1 & -1 \end{pmatrix}$, now

$$U^{-1}AU = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = D.$$

$$\begin{aligned} 3). (1+a) \left(\sum_{n=0}^{2017} (-1)^n \cdot a^n \right) &= \sum_{n=0}^{2017} (-1)^n \cdot a^n + \sum_{n=0}^{2017} (-1)^n \cdot a^{n+1} \\ &= \sum_{n=0}^{2017} (-1)^n \cdot a^n + \sum_{n=1}^{2017} (-1)^{n-1} \cdot a^n \\ &= 1 - a + a^2 - \dots + a^{2017} \\ &\quad + a - a^2 + \dots - a^{2017} + a^{2018} \\ &= 1. \end{aligned}$$

4) $|A_4| = 12$. Suppose that $1 \leq A_4$ is a subgroup of order 6.

Then, $[A_4 : 1] = 2$, thus $1 \trianglelefteq A_4$, Normal. And, $A_4/1 \cong \mathbb{Z}_2$, abelian. Therefore $[A_4, A_4] \leq 1$. Meanwhile since $K = \{1, (12)(34), (14)(23), (13)(24)\} \leq A_4$ and it is Normal; Conjugate preserves order, and every order 2 elements in A_4 are contained in K . Now, $A_4/K \cong \mathbb{Z}_3$, cyclic abelian, thus $A_4' \leq K$. Since A_4 is non-abelian, $A_4' \neq 1$, and since $A_4' \trianglelefteq A_4$, $A_4' = K$.

$$5. \alpha^3 = 5 + 3\sqrt{6}(\sqrt{2} + \sqrt{3}) = 5 + 6\sqrt{3} + 9\sqrt{2}$$

$$\alpha^2 = 5 + 2\sqrt{6}$$

$$\alpha = \sqrt{2} + \sqrt{3}$$

$$\alpha^3 - 6\alpha = 5 + 3\sqrt{2}$$

$$\frac{1}{3}(x^3 - 6x - 5) = r(x) \cdot q$$

6. Let $f(z) = z^{2018} + z^{2017} + 1$. Then, the given integral is

$$\int_{C_R} \frac{zf'(z)}{f(z)} dz = z \log f(z) \Big|_{w_1}^{w_2} - \int_{C_R} \log f(z) dz.$$

#7. Since $e^{-x} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n$ for all $x \in \mathbb{R}$ and $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ in $|x| < 1$,
 $\frac{e^{-x}}{1-x} = \sum_{n=0}^{\infty} C_n x^n$ for all $|x| < 1$, where $C_n = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!} = \sum_{k=0}^n \frac{(-1)^k}{k!}$

Meanwhile, $\frac{e^{-x}}{1-x} \cdot e^x = \frac{1}{1-x}$, so

$$\sum_{n=0}^{\infty} x^n = \left(\sum_{n=0}^{\infty} C_n x^n \right) \cdot \left(\sum_{n=0}^{\infty} \frac{1}{n!} x^n \right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{C_k}{(n-k)!} \right) x^n$$

By the Taylor's Thm, the coefficients must be equal by terms. Thus proved.

2019. fall.

4. Let $D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$ denote the $n \times n$ diagonal matrix.

For any permutation $\sigma \in S_n$, $D_\sigma = \begin{pmatrix} \lambda_{\sigma(1)} & & 0 \\ & \ddots & \\ 0 & & \lambda_{\sigma(n)} \end{pmatrix}$ is similar to D .

Proof. Let $\tau = (i \ j) \in S_n$ ($1 \leq i < j \leq n$) be a transposition.

Put $E_{i,j} = \begin{pmatrix} e_{11} & \dots & e_{jn} \\ & & \\ & & e_{ii} \end{pmatrix}$. Then, by simple calculation, $E_{i,j}^{-1} = E_{i,j}$.

Moreover, for any $A = \begin{pmatrix} \alpha_1 & \dots & \alpha_n \end{pmatrix}$, $AE_{i,j} = \begin{pmatrix} \alpha_{11} & \dots & \alpha_{jn} \\ & & \\ & & \alpha_{ii} \end{pmatrix}$ and similarly $E_{i,j}A = \begin{pmatrix} \beta_{11} \\ \vdots \\ \beta_{jn} \end{pmatrix}$ where $A = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}$.

Therefore, $E_{i,j} \cdot D \cdot E_{i,j} = \begin{pmatrix} \lambda_{\sigma(1)} & & \\ & \ddots & \\ & & \lambda_{\sigma(n)} \end{pmatrix} \sim D$

Finally, since every permutation can be represented by product of transposition, we obtain the result, inductively.

2023, spring.

#6. Suppose that f is continuous on $[0,1]$ satisfying

$$\int_0^1 f(x) \cos x \, dx = 2 \int_0^1 \varphi(x) \, dx$$

for any continuous φ . Find all f .

Proof. Observe that the given equation can be changed as

$$\int_0^1 (f(x) - 2) \cdot \varphi(x) \, dx = 0.$$

So, let $\varphi(x) = f(x) - 2$, then

$$\int_0^1 (f(x) - 2)^2 \, dx = 0.$$

Since $(f(x) - 2)^2 \geq 0$, so $f(x)$ must be the constant, 2.

#17. Let $f: \mathbb{C} \rightarrow \mathbb{C}; z \mapsto u(z) + i v(z)$ be an entire.

If $u^2(z) - 2v^2(z) = 1$ for all $z \in \mathbb{C}$, then f is constant.

Proof. $f^2 = u^2 - v^2 + 2i u \cdot v = 1 + v^2 + 2i u v$

Meanwhile, by the Cauchy-Riemann Equation, $u_x = v_y$ and $v_x = -u_y$.

$$\begin{aligned} \frac{d}{dz} f^2 &= 2v \cdot v_x + 2i \cdot (u_x v + u v_x) \\ &= -2i v \cdot v_y + 2 \cdot (u_y v + u v_y) \end{aligned}$$

$$\text{So, } u_x v + u v_x = -2v v_y$$

Differentiating the given condition:

$$u_x \cdot 2u - 2v_x \cdot v = 0$$

And, by the Cauchy-Riemann,

